INCH-POUND

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DETAIL SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M25A1 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M25A1 Cartridge Actuated Thruster Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550 Ammunition, General Specification For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

11730535	Thruster, Cartridge Actuated, M25A1, Metal Parts Assembly
8797741	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1
8797742	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1
9217579	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1

U.S. Air Force

TPO 00-554-3103	Thruster, Cartridge Actuated, M25A1

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQ Z1.4	Sampling Procedures And Tables For Inspection By Attributes
ANSI/ISO/ASQ A8402	Quality Assurance Terms And Definitions

(Copies of this document are available online at http://e-standards.asq.org/ or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured thruster metal parts assembly shall contain a new shear pin, new O-rings, new piston locking spring, new cap screws, and a new ring.

3.2 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 8,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.3.1).

3.3 Structural strength. The assembled thruster shall withstand a static tensile force of 2,000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.3.1).

3.4 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.4 and 6.5) shall have the existing protective finish stripped and the specified protective finish reapplied.

3.5 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.4.

3.6 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 48 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled with sufficient torque to bottom components.

3.7.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.7.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for thrusters.

3.8 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

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3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections.

a. Quality conformance inspection

4.1.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ISO/ASQ A8402 to define the quality assurance terms used.

4.1.2 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a.Temperature: room ambient (68 to 77°F)
b.Altitude: sea level
c.Humidity: 50 percent relative, maximum
d.Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Piston, Drawing 10534181, covering a detail of Drawing 11730535.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
Minor		
201	Depth of threaded holes	Gage
202	Radius or chamfer incorrect	Gage
203	Radius or chamfer missing	Visual
204	Surface finish improper	Visual

4.3.2.3 Body Assembly, Drawing 10534179, covering a detail of Drawing 11730535.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	True position of slots	Gage
105	Symmetry of slots with each other and with body datum diameter	Gage
106	Depth of slots	Gage
107	Width of slots	Gage
108	Deleted	
Minor		
201	Radius or chamfer incorrect	Visual
202	Surface finish improper, except as otherwise classified	Visual

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4.3.2.4 Guide, Retainer Assembly, Drawing 10521616, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
106	Pellets loose, missing, or improperly inserted	Visual

Minor
None defined

4.3.2.5 Key, Drawing 10534180, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual

Minor
None defined

4.3.2.6 Sleeve Assembly, Drawing 10521617, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual
106	Deleted	
107	Deleted	
108	Deleted	
109	Deleted	
110	Deleted	

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Minor

201	Surface finish improper, except as otherwise classified	Visual

4.3.2.7 Trunnion, Drawing 8594482, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Diameter of lugs	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual

Minor
None defined

4.3.2.8 Ring, Drawing 10534183, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Concentricity of exterior diameter with interior diameter	Gage
102	Concentricity of small holes	Gage
103	Concentricity of center bore with small holes	Gage
104	Chamfers at exterior and interior surfaces	Visual
105	Protective finish inadequate or defective	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual

Minor

201	Diameter at exterior surface	Gage
202	Diameter at interior surface	Gage
203	Diameter of holes	Gage
204	Diameter of counterbore	Gage
205	Depth of counterbore	Gage
206	Length from front face to rear face	Gage
207	Location of holes	Gage
208	Radius or chamfer incorrect	Gage
209	Radius or chamfer missing	Visual
210	Surface finish improper	Visual
211	Marking missing, incorrect, or illegible	Visual

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4.3.2.9 Piston, Unlocking, Drawing 10534182, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Chamfer improper or incorrect	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual

Minor		
201	Radius or chamfer missing except as specified	Visual
202	Marking missing, incorrect, or illegible	Visual
203	Surface finish improper	Visual

4.3.2.10 Head, Drawing 10521455, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Thread damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual

Minor
None defined

4.3.2.11 Pin, Firing, Drawing 8593864, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual

Minor
None defined

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4.3.2.12 Assembly, Drawing 11730535, less all components, except Drawings 10521616, 8593864, and MS29513-12.

Critical		
1	Firing pin protrusion	Gage
Major		
101	Any part missing	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
None defined		

4.3.2.13 Assembly, Drawing 11730535.

Critical		
None defined		
Major		
101	Trunnion not free to rotate 360 degrees	Manual
102	Any part missing	Visual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Foreign matter	Visual
Minor		
201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.3.2.14 Spring, Piston Locking, Drawing 10534184, covering a detail of Drawing 11730535.

Critical		
None defined		
Major		
101	Load at specified heights, incorrect	Gage
102	Solid length, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter	Visual
Minor		
201	Outside diameter	Gage

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4.3.2.15 Pin, Shear, Drawing 10521618, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Surface finish improper	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Profile of end	Gage

4.3.2.16 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797741.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.3.2.17 Box, Packing, Ammunition, Drawing 8797742.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual

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205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.18 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217579.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loosed, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each body and head shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.2 shall be cause for rejection of the subassembly.

4.4.2 Structural strength. Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.3 shall be cause for rejection of the thruster.

4.4.3 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.4 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.4. Failure of any unit in the inspection sample to comply with 3.5 shall be cause for rejection of the assembly.

4.4.5 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to application of protective finish the body and head shall be mounted in a device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

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4.5.2 Structural strength. The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.3. The test shall be performed in a fixture in conjunction with a tensile testing machine or other device.

4.5.3 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.5 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2 and 6.7). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a.Title, number, and date of this specification

b. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2)

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c.The responsible inspection activity (see 4.1)

d.Any special instruction requirements (see 4.3)

e.Level of packing and marking (see 5.1 and 6.7.1)

6.3 Definition.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.4 Exterior components. Remanufactured exterior components are: body, head, and trunnion.

6.5 Reclaimed components. Reclaimed components, which include piston, sleeve, ring, and unlocking piston with damaged protective finish, should be reprocessed by being stripped of protective finish and replated in accordance with the applicable drawings and specifications.

6.6 Old engraved or stamped lot and serial numbers. Old serial numbers and lot numbers engraved or stamped on metal parts should be over-stamped by x's to obliterate those markings before application of final protective finish.

6.7 Packaging and packing.

6.7.1 Packing and marking. Packing and marking should be Level A or B, as specified by the contracting activity (see 6.2). Level A should be in accordance with Drawings 8797741 and 8797742. Level B should be in accordance with Drawings 8797741 and 9217579.

6.7.2 Air Force packaging. Air Force packaging should be in accordance with TPO 00-554-3103.

6.8 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.9 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy - OS
Air Force - 99

Preparing activity
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